

Vehicle Overweight Detection System

Project Overview

The **Vehicle Overweight Detection System** is an Arduino Uno based embedded system designed to monitor the weight of a vehicle and indicate whether the measured load is within the normal range or exceeds the defined limits.

The system uses an **HX711 load-cell amplifier** to measure weight. The measured value is processed by the Arduino Uno and displayed on a **16×2 I2C LCD**. LEDs provide visual indications of the weight condition, while a buzzer gives an audible warning when the vehicle is detected as heavily overloaded.

The project is developed and simulated using **Wokwi**.

Objective

The main objective of this project is to develop a simple and cost-effective system for detecting vehicle overloading.

The system:

- **Measures the vehicle/load weight.**
- **Displays the measured weight on an LCD.**
- **Classifies the weight into different levels.**
- **Provides visual indication using LEDs.**
- **Activates a buzzer when the weight reaches the high/overweight condition.**

Components Used

Component	Purpose
Arduino Uno	Main microcontroller
HX711 Load Cell Module	Weight measurement interface
Load Cell	Measures the applied load
16×2 I2C LCD	Displays weight and status
Green LED	Indicates normal weight
Yellow LED	Indicates medium weight
Red LED	Indicates high weight
Buzzer	Provides overload warning

Pin Connections

HX711

HX711 Pin	Arduino Uno
DT	D2
SCK	D3
VCC	5V
GND	GND

LEDs

Component	Arduino Pin
Green LED	D6
Yellow LED	D5
Red LED	D4

Buzzer

Buzzer	Arduino Uno
Positive	D7
Negative	GND

I2C LCD

LCD Pin	Arduino Uno
SDA	A4
SCL	A5
VCC	5V
GND	GND

Working Principle

The system works in the following sequence:

1. The Arduino Uno initializes the LCD, LEDs, buzzer, Serial communication, and HX711 module.
2. The HX711 reads the signal from the load cell.
3. The Arduino obtains the load-cell reading.
4. The reading is converted into a weight value using the configured scale factor.
5. The measured weight is displayed on the LCD.
6. The Arduino compares the measured weight with predefined thresholds.
7. The corresponding LED is switched ON.

8. When the weight reaches the high-weight condition, the red LED and buzzer are activated.
 9. The LCD displays the current status.
 10. The process repeats continuously.
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Weight Status Indication

The current program uses the following conditions:

Weight Condition	Status	Indication
<15	NORMAL	Green LED
15–<40	MEDIUM	Yellow LED
>41	HIGH	Red LED + Buzzer

Note: The exact weight thresholds and calibration factor can be modified in the Arduino program according to the actual load cell and application requirements.

Software Used

- Arduino IDE
- Embedded C/C++
- Wokwi Simulator

Libraries Used

- LiquidCrystal_I2C
- HX711

Program Logic

The important part of the program continuously checks whether the HX711 is ready:

```
if (scale.is_ready()) {  
    reading = scale.read();  
    float weight = reading / 420;  
}
```

The weight is then compared with the predefined limits:

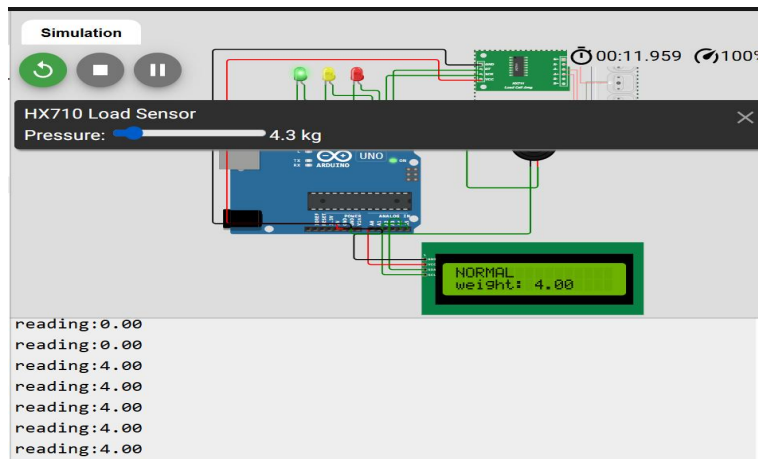
```
if (weight < 15) {  
    digitalWrite(GLED, HIGH);  
    lcd.print("NORMAL");  
}  
else if (weight < 40) {  
    digitalWrite(YLED, HIGH);  
    lcd.print("MEDIUM");  
}
```

```
else if (weight > 41) {  
    digitalWrite(RLED, HIGH);  
    digitalWrite(BUZZER, HIGH);  
    lcd.print("HIGH");  
}
```

Output

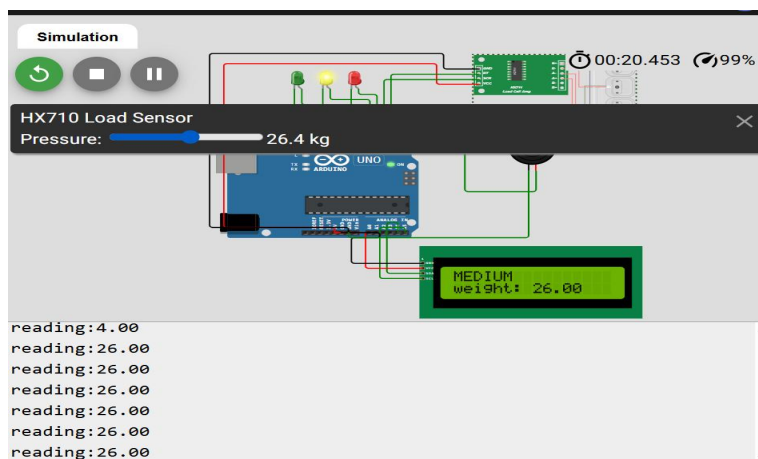
The LCD displays information similar to:

NORMAL
weight: 10



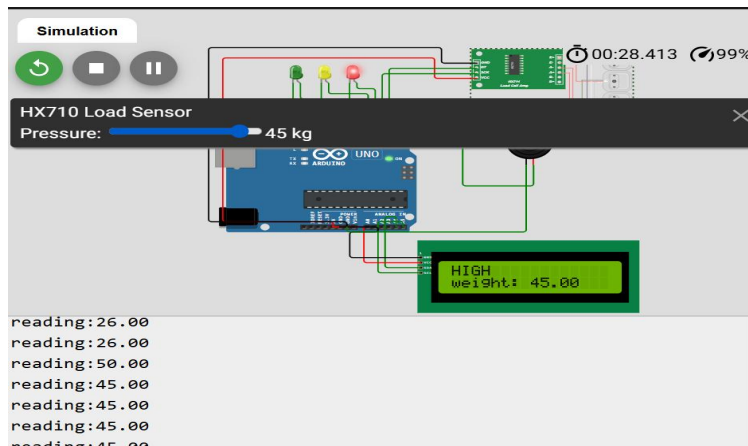
For a medium load:

MEDIUM
weight: 25



For a high load:

HIGH
weight: 45



During the high-weight condition:

- **Red LED turns ON.**
- **Buzzer turns ON.**
- **LCD displays HIGH.**

Simulation

This project can be simulated using **Wokwi**, which allows the Arduino circuit and program to be tested without physical hardware.

The project contains:

- **sketch.ino** – Arduino source code
- **diagram.json** – Wokwi circuit configuration
- **libraries.txt** – Required libraries
- **wokwi-project.txt** – Wokwi project information

Project Structure

```
VEHICLE-OVER-WEIGHT-DETECTION-SYSTEM/  
├── sketch.ino  
├── diagram.json  
├── libraries.txt  
└── wokwi-project.txt
```

Future Enhancements

The project can be improved by adding:

- **IoT-based weight monitoring.**

- **Cloud data storage.**
 - **GSM/Wi-Fi notifications.**
 - **Automatic vehicle gate control.**
 - **Real-time mobile application monitoring.**
 - **Data logging for multiple vehicles.**
 - **More accurate load-cell calibration.**
 - **Automatic number-plate recognition.**
 - **Automatic penalty/alert generation for overloaded vehicles.**
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Skills Demonstrated

This project demonstrates practical knowledge of:

- **Embedded C/C++**
 - **Arduino programming**
 - **Sensor interfacing**
 - **HX711 load-cell interfacing**
 - **I2C communication**
 - **LCD interfacing**
 - **Digital I/O**
 - **Embedded system logic**
 - **Hardware simulation using Wokwi**
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Author

Sony Myadari

Embedded Systems / Software Engineering Enthusiast

License

This project is intended for educational and demonstration purposes.